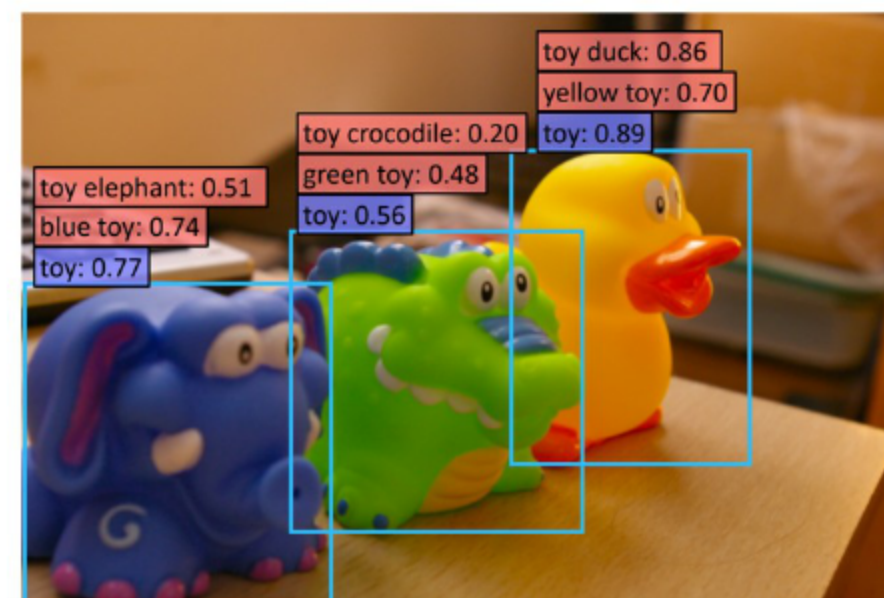


Open-Vocabulary Object Detection via Vision and Language Knowledge Distillation (ViLD)

Xiuye Gu, Tsung-Yi Lin, Weicheng Kuo, Yin Cui

01 Why Open-Vocabulary Detection?



- Conventional detectors struggle with long-tailed categories due to costly box/mask annotations.
- Goal: detect objects from arbitrary text queries, including unseen categories.

01 Key Challenge: Data Scarcity And Long Tail

- Datasets exhibit Zipf-like distributions; many categories have few examples.
- Manual expansion via bounding-box labels is infeasible at scale.
- Transferring web-scale image-text recognition to object-level detection is hard.

02 Background And Related Work

Zero-Shot And Large-Vocabulary Recognition

Early zero-shot used attributes, hierarchies, and word embeddings.

CLIP/ALIGN learn joint embeddings from large image-text corpora.

These focus on whole-image classification, not detection.

Open-Vocabulary And Zero-Shot Detection

Prior methods align region features to text or pretrain with captions.

They use small vocabularies and train supervised detection.

This work distills from strong image-text classifiers into two-stage detectors.

03 Key Contributions

ViLD: Distilling Vision-Language Knowledge

Distill teacher text and image embeddings into detector region embeddings.

Enables open-vocabulary classification with base-only supervision.

Simple, scalable training compatible with two-stage detectors.

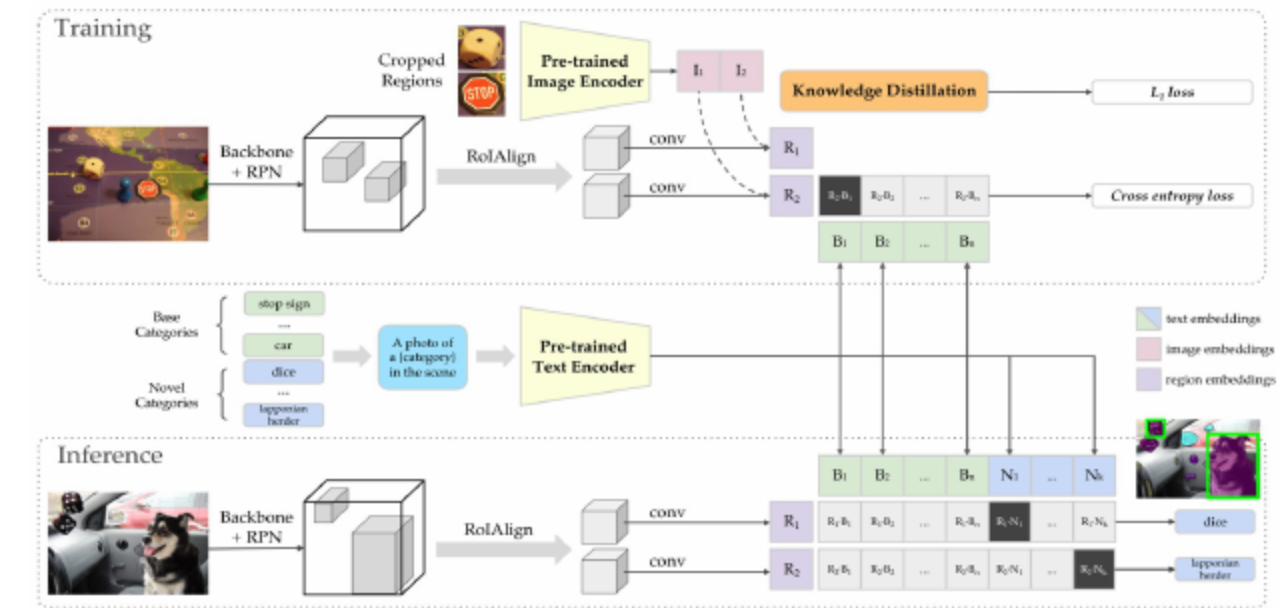
Efficient And Transferable Open-Vocabulary Detection

Single pass region embeddings replace slow proposal-wise classification.

Supports various teachers and backbones; class-agnostic localization.

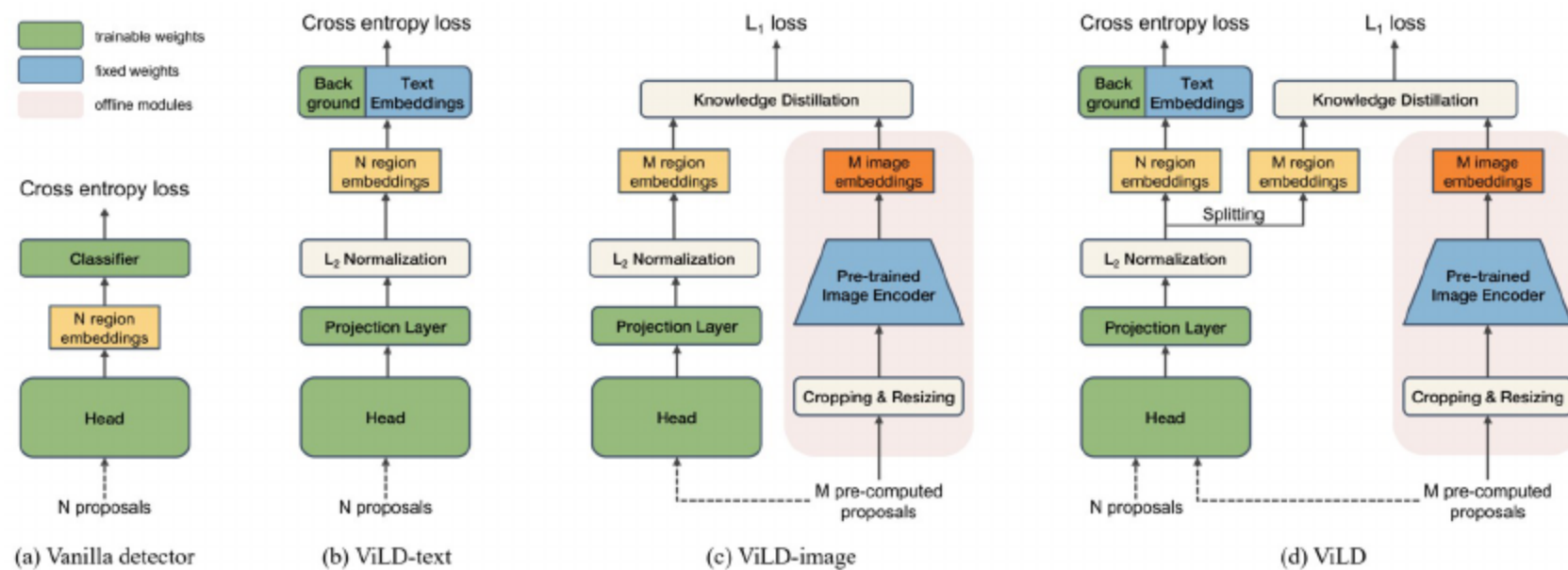
Zero-shot transfer by swapping text embeddings only.

04 Vocabulary Classification



- Class-agnostic box/mask heads generate proposals that generalize to novel objects.
- Classify regions via similarity to teacher text embeddings.
- Recognize arbitrary text-specified categories at inference.

04 Teacher-Student Distillation Setup

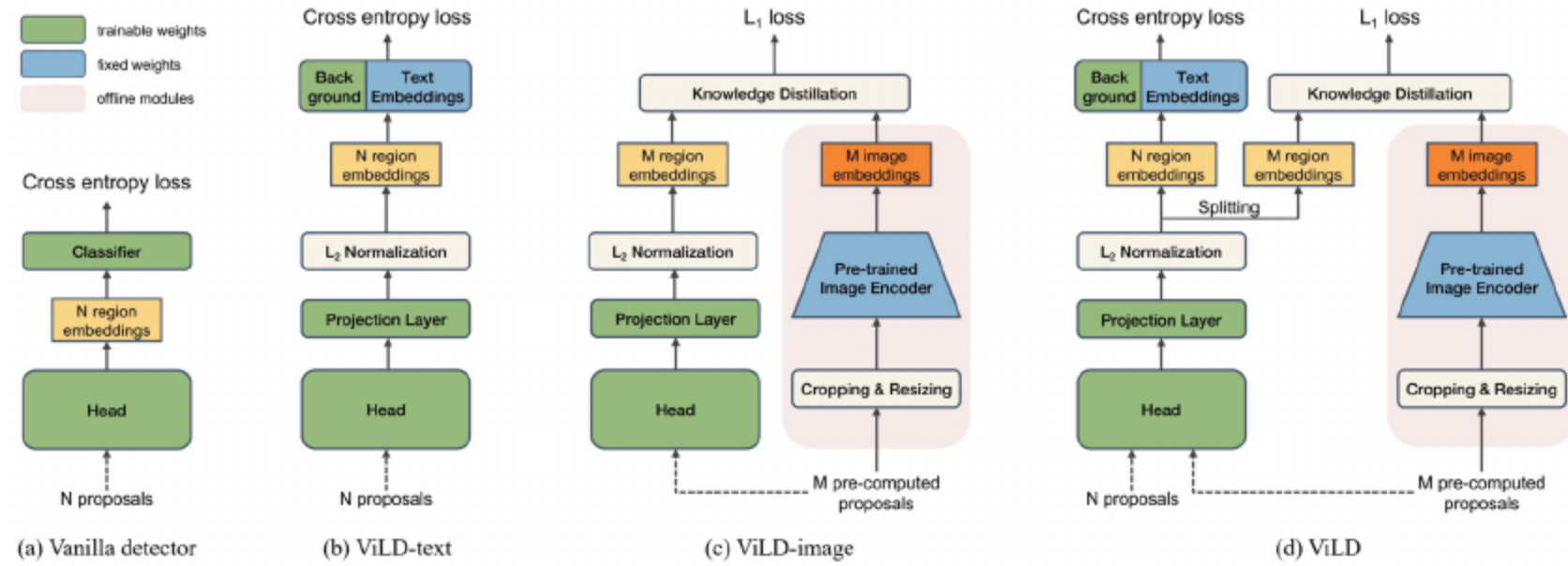


- Teacher supplies text embeddings for prompts and image embeddings for proposal crops.
- Student detector learns region embeddings compatible with teacher outputs.
- Loss: text classification plus L₁ image embedding distillation.

05 Step 1: Class-Agnostic Localization

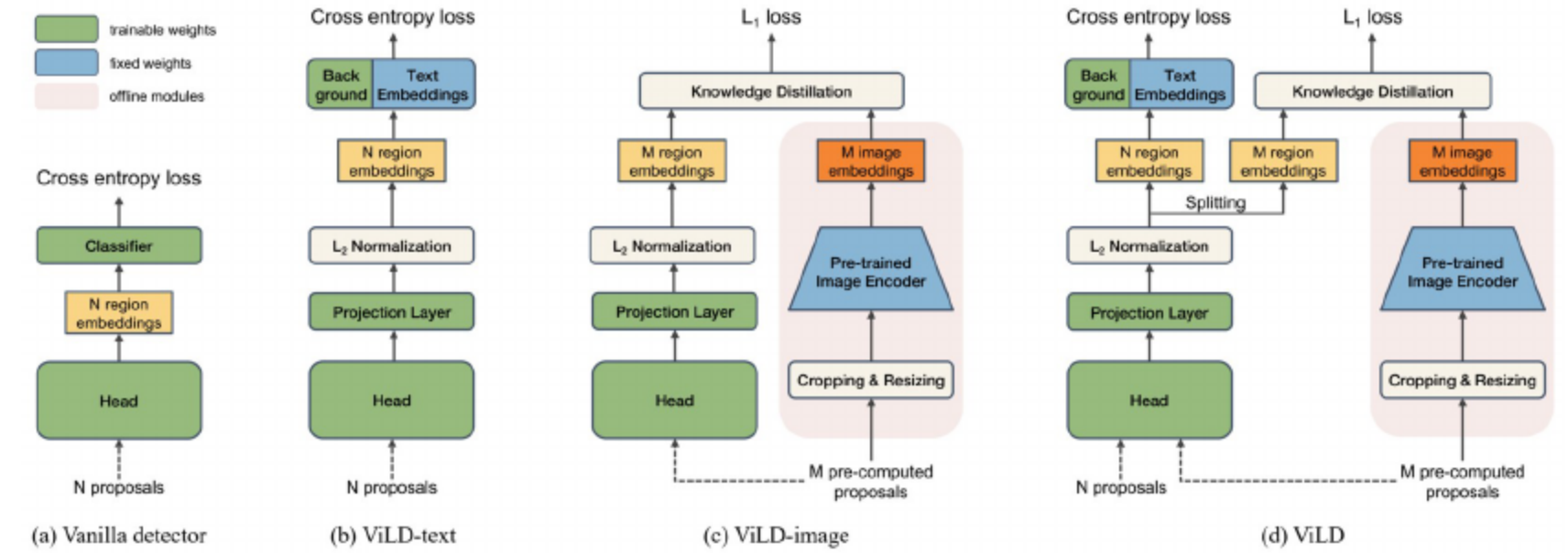
- Replace class-specific heads with class-agnostic box and mask predictors.
- Trained on base categories; proposals generalize with minimal recall drop.
- Provides candidate regions for open-vocabulary classification.

05 Embeddings)



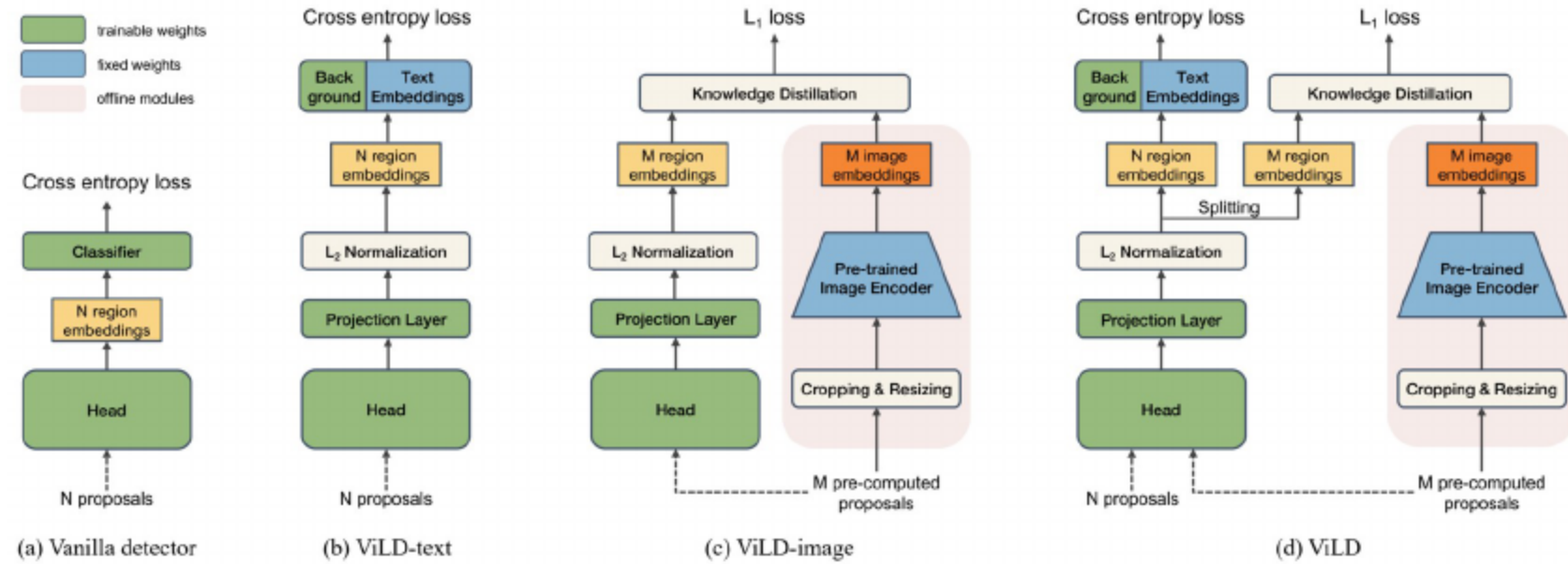
- Use multiple prompt templates to generate base-category text embeddings.
- Replace classifier with fixed text embeddings plus learnable background.
- Train with temperature-scaled cosine similarity cross-entropy.

05 Step 3: ViLD-Image (Distill Image Embeddings)



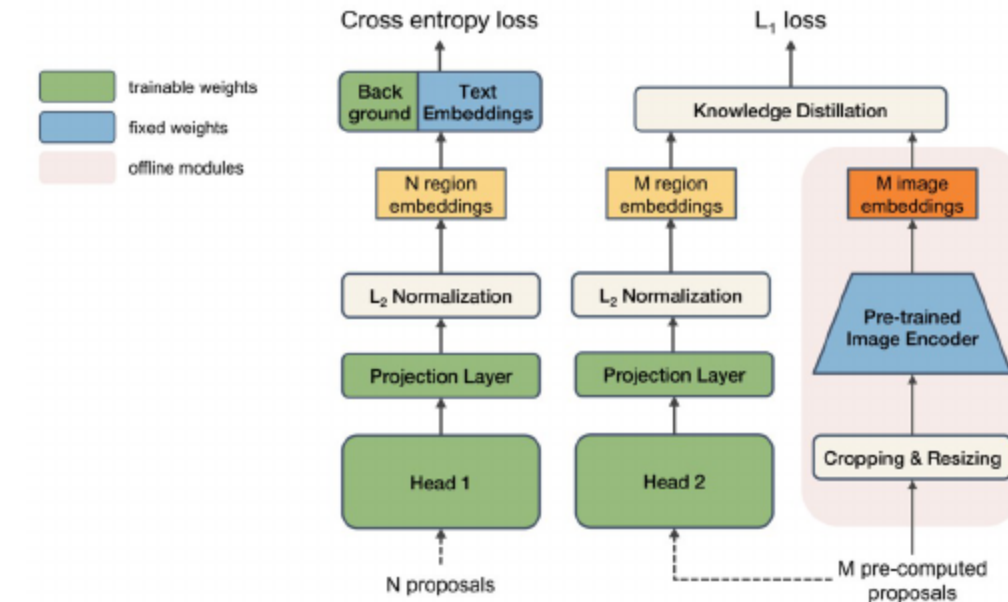
- Collect proposals; compute ensembled teacher embeddings from 1×/1.5× crops.
- Train region embeddings to match teacher via L1 loss.
- Leverages unlabeled proposals to absorb novel visual knowledge.

05 Step 4: Joint Objective And Inference



- Combine text classification and image L1 losses with weight w .
- Inference uses text embeddings as the classifier over regions.
- No teacher computation at test time.

05 Model Ensembling Strategies



- ViLD-text+Teacher: ensemble proposal scores with teacher; slower.
- ViLD-ensemble: separate heads; ensemble via weighted geometric mean.
- Reduces objective contention with similar runtime.



- LVIS v1: 866 base, 337 novel; AP_r as key metric.
- COCO zero-shot: 48 base, 17 novel; AP at IoU 0.5.
- Transfers to VOC, COCO, Objects365 without finetuning.

- Mask R-CNN with ResNet-FPN; 1024×1024 inputs; large-scale jittering.
- Teachers: CLIP (ViT-B/32, ViT-L/14), ALIGN; $\tau=0.01$.
- Max 300 detections; normalize ensembled crop embeddings.

Supervision	AR _r @100	AR _r @300	AR _r @1000
base	39.3	48.3	55.6
base + novel	41.1	50.9	57.0

- RPN trained on base-only attains similar AR on novel categories.
- Approximately 2 AR gap versus base+novel at 100–1000 proposals.
- Validates class-agnostic proposals for open-vocabulary detection.

Method	AP _r	AP _c	AP _f	AP
Supervised (base class only)	0.0	22.6	32.4	22.5
CLIP on cropped regions w/o objectness	13.0	10.6	6.0	9.2
CLIP on cropped regions	18.9	18.8	16.0	17.7
Supervised (base+novel)	4.1	23.5	33.2	23.9
Supervised-RFS (base+novel)	12.3	24.3	32.4	25.4

- CLIP on proposal crops yields strong AP_r but slower inference.
- Ensembling with objectness boosts base and novel AP.

07 ViLD Variants On LVIS

Method	AP_r	AP_c	AP_f	AP
Supervised (base class only)	0.0	22.6	32.4	22.5
CLIP on cropped regions w/o objectness	13.0	10.6	6.0	9.2
CLIP on cropped regions	18.9	18.8	16.0	17.7
Supervised (base+novel)	4.1	23.5	33.2	23.9
Supervised-RFS (base+novel)	12.3	24.3	32.4	25.4

- ViLD-text surpasses GloVe; ViLD-image achieves label-free gains.
- Combined ViLD improves AP_r substantially over supervised baselines.
- Ensembling separate heads further boosts AP_r and overall AP.

07 COCO Comparisons

Method	Training source	Novel AP	Base AP	Overall AP
Bilen & Vedaldi (2016)	image-level labels in $C_B \cup C_N$	19.7	19.6	19.6
Ye et al. (2019)		20.3	20.1	20.1
Bansal et al. (2018)	instance-level labels in C_B	0.31	29.2	24.9
Zhu et al. (2020)		3.41	13.8	13.0
Rahman et al. (2020)		4.12	35.9	27.9
Zareian et al. (2021)	image captions in $C_B \cup C_N$ instance-level labels in C_B	22.8	46.0	39.9
CLIP on cropped regions	image-text pairs from Internet (may contain $C_B \cup C_N$)	26.3	28.3	27.8
ViLD-text		5.9	61.8	47.2
ViLD-image		24.1	34.2	31.6
ViLD ($w = 0.5$)		27.6	59.5	51.3

- ViLD achieves 27.6 Novel AP and 51.3 overall AP on COCO.
- Outperforms prior zero-shot detection methods by large margins.

07 Scaling Teachers And State Of The Art

Method	AP_r	AP_c	AP_f	AP
CLIP on cropped regions	19.5	19.7	17.0	18.6
ALIGN on cropped regions	39.6	32.6	26.3	31.4

- Performance scales with stronger teachers (CLIP ViT-L/14, ALIGN).
- ALIGN-based variants approach LVIS Challenge winner performance.

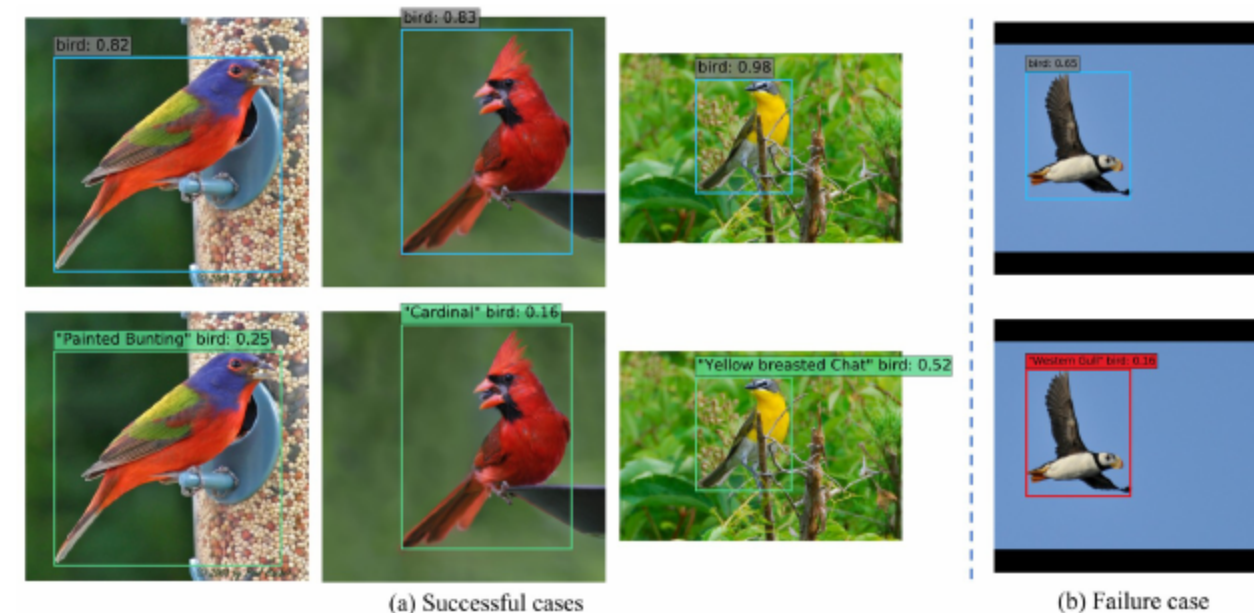
07 Zero-Shot Transfer To New Datasets

Method	PASCAL VOC [†]		AP	COCO		AP	Objects365	
	AP_{50}	AP_{75}		AP_{50}	AP_{75}		AP_{50}	AP_{75}
ViLD-text	40.5	31.6	28.8	43.4	31.4	10.4	15.8	11.1
ViLD	72.2	56.7	36.6	55.6	39.8	11.8	18.2	12.6
Finetuning	78.9	60.3	39.1	59.8	42.4	15.2	23.9	16.2
Supervised	78.5	49.0	46.5	67.6	50.9	25.6	38.6	28.0

- Transfers to VOC, COCO, Objects365 by swapping text embeddings.
- Strong zero-shot AP despite no finetuning.

Distill loss	Distill weight w	AP_r	AP_c	AP_f	AP
No distill	0.0	10.4	22.9	31.3	24.0
	0.5	13.7	21.7	31.2	24.0
	1.0	12.4	22.7	31.4	24.3
	2.0	13.4	22.0	30.9	24.0
$\mathcal{L}_1$ loss	0.05	12.9	22.4	31.7	24.4
	0.1	14.0	20.9	31.2	23.8
	0.5	16.3	19.2	27.3	21.9
	1.0	17.3	18.2	25.1	20.7

- Higher image-distillation weight improves novel AP, hurts frequent/common.
- Separate-head ensembling reduces competition and boosts both.

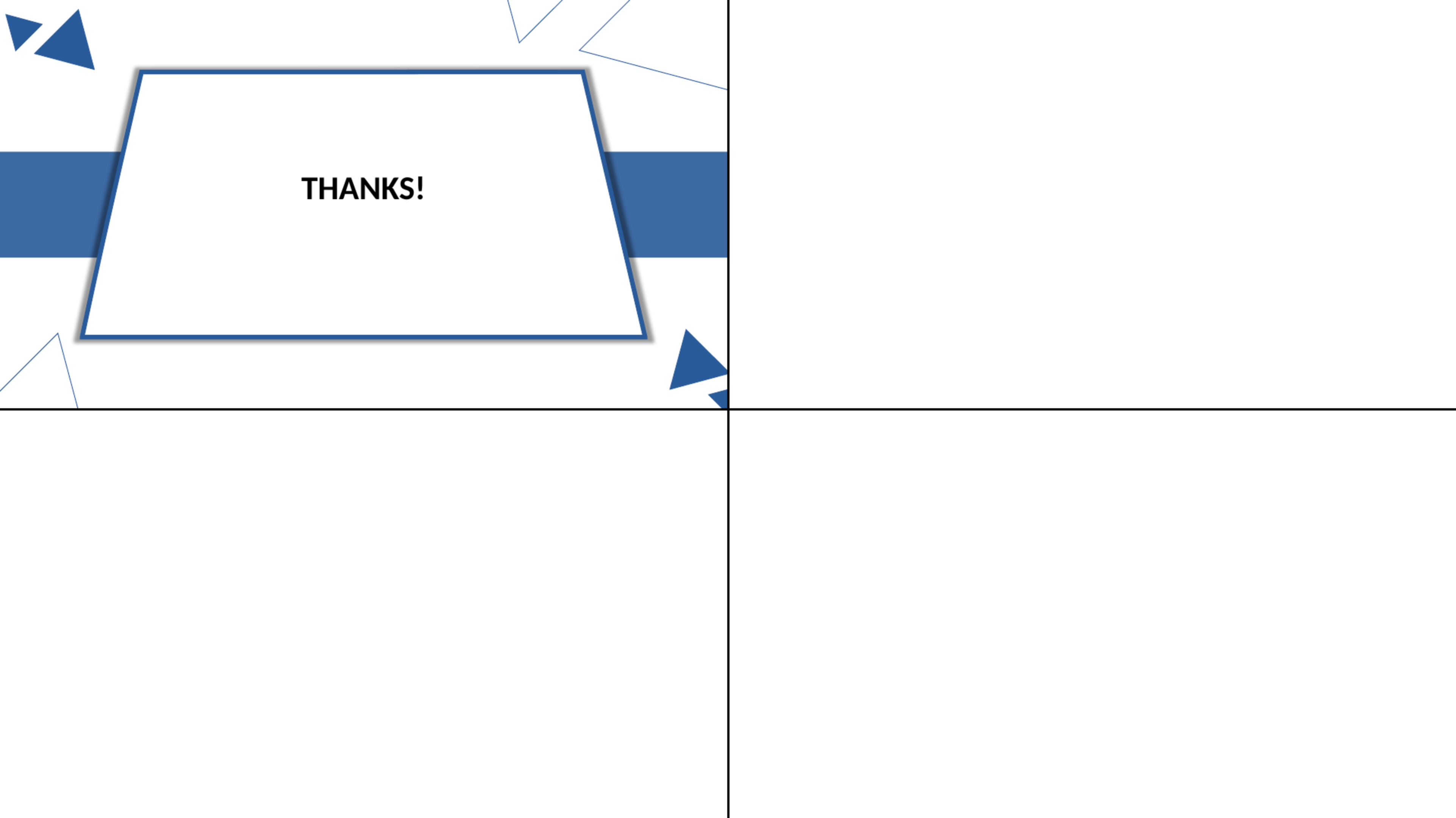


- Region embeddings enable fine-grained and attribute queries on-the-fly.
- Composing categories with attributes expands vocabularies (e.g., colors, bird species).



- ViLD distills vision-language knowledge for efficient open-vocabulary detection.
- State-of-the-art novel-category performance; scales with stronger teachers.
- Transfers across datasets without finetuning.

- Improve proposals for unseen objects and balance text-image objectives.
- Refine ensembling strategies for robust performance.
- Consider ethical impacts; ensure responsible deployment in sensitive domains.

The background features a white canvas with a thin black horizontal line across the middle and a thin black vertical line on the right side. In the upper-left quadrant, there are several blue geometric elements: a small triangle, a larger triangle, a horizontal rectangle, and a parallelogram. The word "THANKS!" is centered within this parallelogram. In the upper-right quadrant, there are thin blue lines forming a larger triangle and a smaller triangle. In the lower-left quadrant, there is a thin blue line forming a triangle. In the lower-right quadrant, there is a blue triangle pointing towards the center.

THANKS!